Muhammad Bilal Javaid

Date: Monday, May 19, 2025

Valid until: Tuesday, May 20, 2025

Development of Smart Health Band

Client	ABC		
Estimated effort	6 Weeks		
Estimated cost (excluding VAT if applicable)	Full price (@00/hr)	$ 000	$ 000

(*: This cost includes only labor expenses and does not account for the cost of components, transportation charges, or any other additional expenses.)

Terms as per standard Algorithm Coding Experts Client Agreement, where this document is the indication of estimated effort referred to in 2.2.1.
The final word on any estimate or proposal will be up to our team of specialists once they have had the opportunity to review and discuss your requirements with you in detail.

Project Overview

This project aims to design and prototype an ultra-slim, screenless smart health band that meets medical-grade accuracy and 2+ days battery life, all within a 5.5 mm thickness envelope. The device will integrate BLE 5.0 connectivity, high-accuracy biometric sensors, gesture detection, and haptic feedback. The firmware will be optimized for power efficiency and OTA updates. Future expansion includes a smart dock with display and charging features. This feasibility report evaluates the technical roadmap, hardware stack, and integration plan for the base unit, covering end-to-end development through 1–2 physical prototypes.

Key Requirements

- Use medical-grade biometric sensors.
- Maximum thickness: 5.5 mm (target 4.0–5.0 mm for internal stack).
- BLE 5.0 sync with mobile app.
- At least 2+ days of battery life.
- Tap and double-tap gesture detection.
- No screen or buttons; haptic feedback only.
- OTA firmware updates via BLE.
- Charging via 2 pogo pins.
- UART dock interface via 2 pogo pins.
- Waterproofing: IP67 minimum.
- Support future dock expansion.

This report covers the feasibility of optimizing the system, along with a road-map for refinement and implementation.

Feasibility Analysis

Our Approach

To fulfill the requirements outlined for this project, our proposed approach is detailed as follows:

1. PCB Design & Layout Optimization

We will design a custom 2-layer or 4-layer PCB in KiCad, focused on a compact, stacked layout. All components will be carefully placed to meet the 4.0–5.0 mm internal height limit.

- Use KiCad for schematic and PCB design.
- Maintain PCB footprint within 42mm × 22mm.
- Stack low-profile components (≤1.0 mm height).
- Follow DFM-friendly layout practices for enclosure fit.

2. MCU & BLE Connectivity

We will integrate the nRF52840 for BLE 5.0 and ultra-low power processing. This SoC offers excellent support for BLE, deep sleep, and peripheral interfacing.

- Use nRF52840 with integrated BLE 5.0 and OTA support.
- Enable UART and I^2C for sensor and dock communication.
- Configure custom GATT profile for app sync.
- Support OTA firmware upgrades over BLE.

3. Medical-Grade Sensor Integration

We will use best-in-class sensors for HR, SpO_2, temperature, and motion. Each sensor is chosen for accuracy, low power, and compact form factor.

- MAX86141 + MAX32664 for HR, HRV, SpO_2, PTT.
- TMP117 for body and skin temperature.
- LSM6DS3 or BMI270 for motion and gesture detection.
- Interface sensors over I^2C to nRF52840.

4. Power System & Battery Life

We will optimize power paths and firmware logic to ensure 2–3+ days runtime. A 125–200 mAh LiPo battery with PCM protection will be used.

- Use 125–200 mAh LiPo cell with PCM.
- Optimize sensor sampling and BLE intervals.
- Enter deep sleep when idle or removed.
- Monitor battery level and report via BLE.

5. Mechanical Constraints & Component Selection

Component heights and placement will be selected to stay within the target thickness. Shielding, stack height, and thermal considerations will be validated.

- Choose components ≤1.0 mm where possible.
- Use flat flex or edge-mount connectors if needed.
- Match layout to 3D enclosure draft from designer.

6. Gesture, Haptics, and Wear Detection

The IMU will be used for tap/double-tap gestures and wear detection. A flat vibration motor will provide silent haptic alerts.

- Program IMU for tap/double-tap detection.
- Enable wear detection via motion and skin temperature.
- Add ultra-thin vibration motor for alerts.
- Wake MCU from sleep based on movement.

7. Firmware Development & OTA

We will develop embedded firmware in C/C++ using the Nordic SDK. The firmware will handle sensor readouts, BLE sync, power modes, and OTA updates.

- Use Nordic SDK for firmware development.

- Implement real-time and processed data reporting.
- Enable secure OTA firmware updates.
- Debug using SWD and UART logging.

8. Pogo Pin Charging & Dock Interface

The base unit will use 4 pogo pins: 2 for charging and 2 for UART to the future dock. These will be placed for mechanical alignment and future dock attachment.

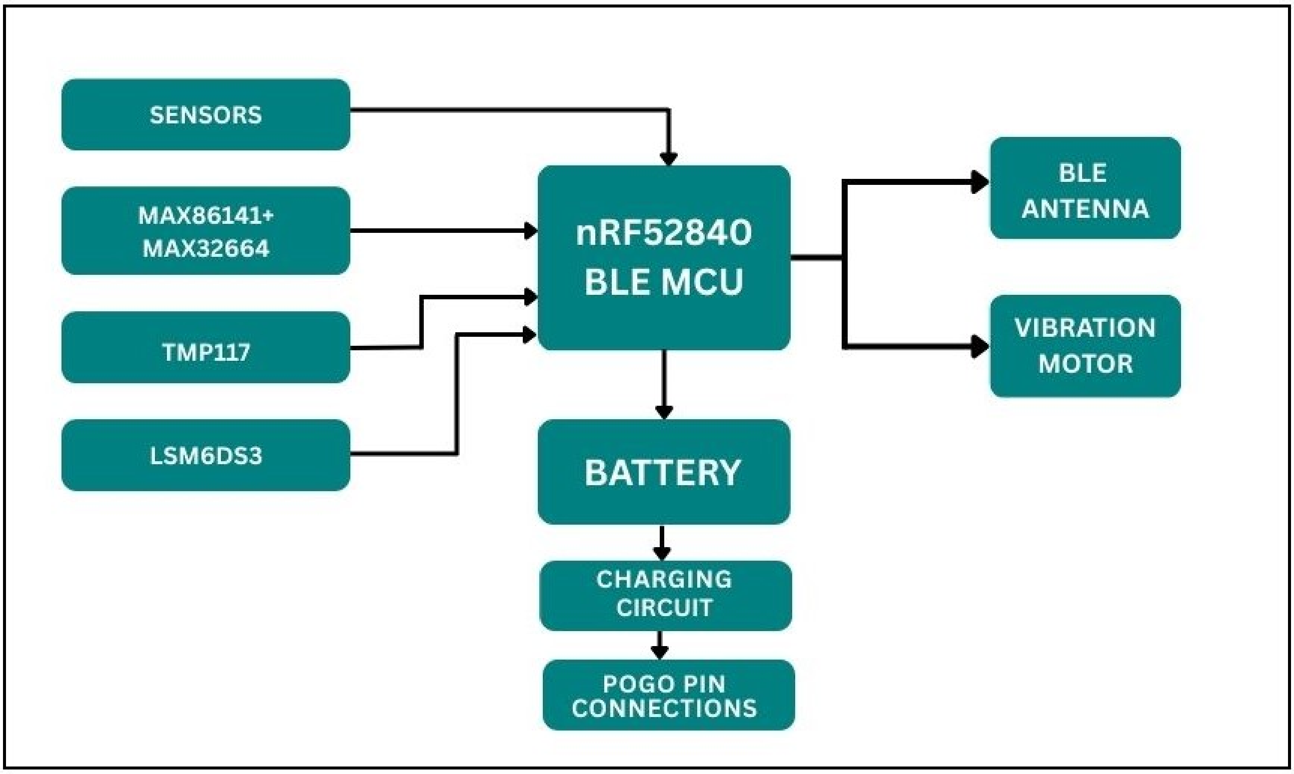
- Define 4-pin pogo layout (GND, VCC, TX, RX).
- Design safe charging circuit with ESD protection.
- Ensure stable mechanical placement for docking.
- UART interface will be idle unless dock is attached.

9. Prototyping & Testing

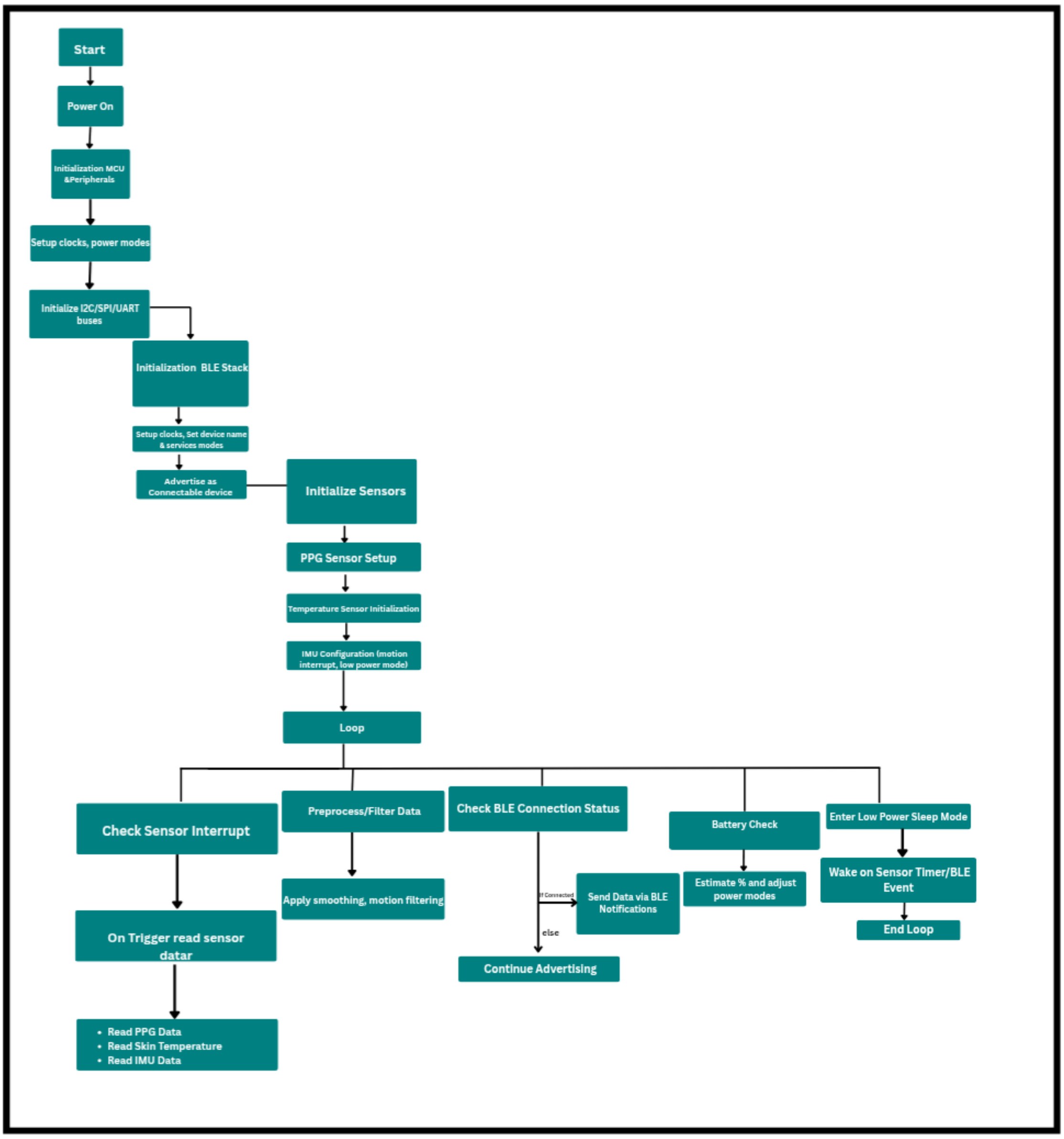
We will build and test 1–2 complete functional prototypes. All features BLE, sensors, battery life, haptics will be validated in hardware.

- Manufacture PCB (JLCPCB or similar).
- Assemble using SMT and reflow process.
- Validate BLE sync, sensor accuracy, haptics.
- Test charging and power management.

Block Diagram



Process Flow Chart



Discussion Points

Query 01:

What strategies or layout options could help us achieve the internal thickness target of 4.0–5.0 mm?

Our Response:

Use 0.8mm or 0.6mm thickness option for PCB. Use smaller package components such as 0201 to keep the height minimum but do not go less than that because most PCB manufacturers cannot assemble components smaller than 0201. Try to route the PCB in 2 layers or maximum 4 layers because more no of layers can increase the thickness of the PCB.

Query 02:

Do you have suggestions for alternative sensors or architecture that preserve our top 3 priorities: medical-grade accuracy, ultra-slim design, and battery life?

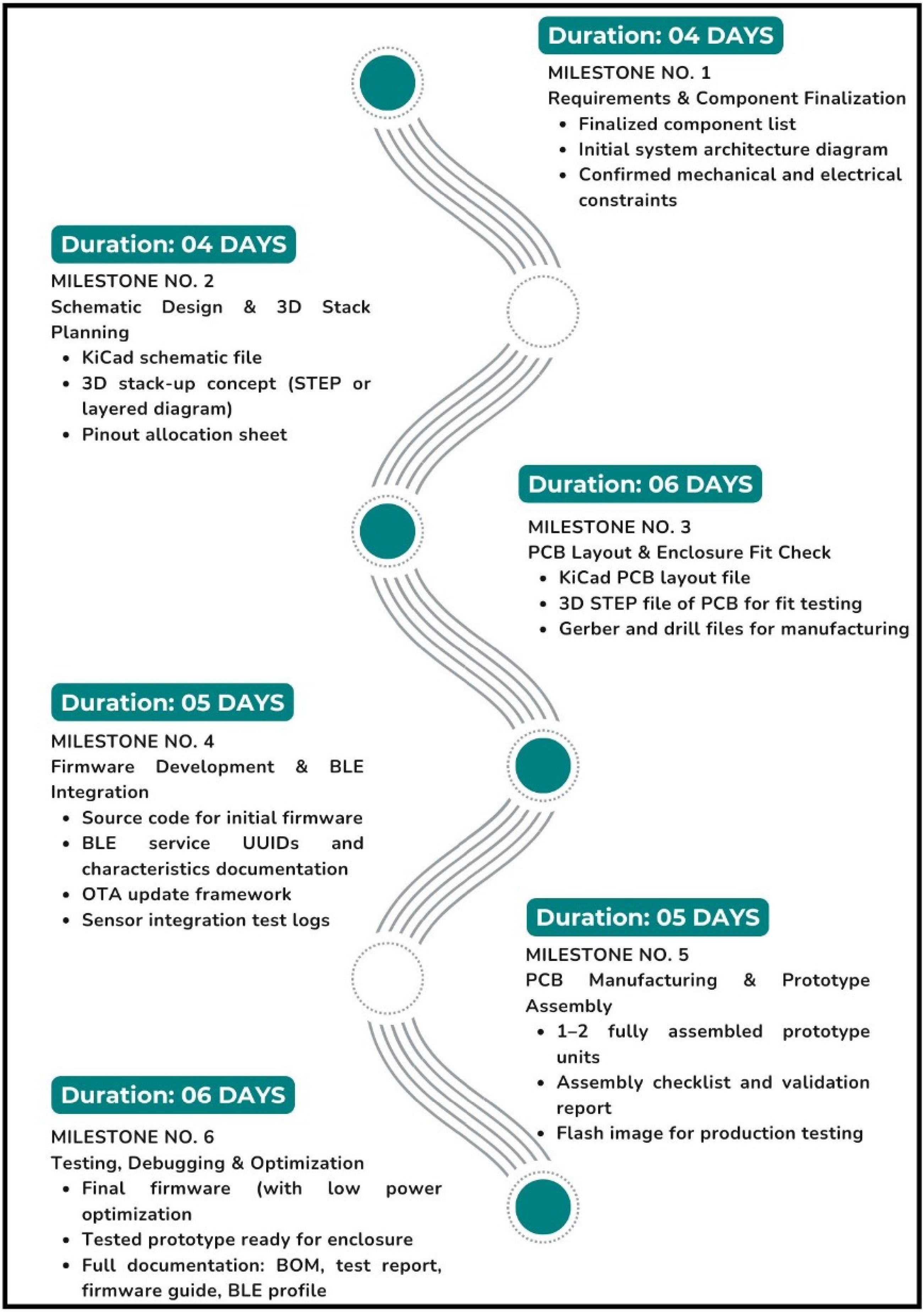
Our Response:

Better Sensor suggested for PPG (HR/HRV/SpO_2/PTT) is AS7038RB or AS7050 because they come in small package of 1.65 mm thick, 3.7×3.1 mm and have low power consumption than MAX86141 + MAX32664 combo. For temperature sensing we would recommend MLX90632 which is thinner.

Conclusion

The proposed ultra-slim health band is technically feasible using available components and design techniques. Key challenges such as height constraints, power optimization, and medical-grade accuracy can be addressed through careful component selection and firmware design. The modular approach also sets a strong foundation for future dock integration.

Milestones and Expected Timeline



Milestone 1: Requirements & Component Finalization

Duration: 4 Days

Activities:

- Review client specs and feasibility constraints
- Finalize sensor selection: MAX86141, TMP117, BMI270
- Choose MCU (nRF52840) and compatible battery

Deliverables:

- Finalized component list
- Initial system architecture diagram
- Confirmed mechanical and electrical constraints

Milestone 2: Schematic Design & 3D Stack Planning

Duration: 4 Days

Activities:

- Design schematic using KiCad
- Plan I^2C, UART, and GPIO lines for sensor & motor integration
- Model internal component stack for ≤5.0 mm height
- Validate pogo pin functions and interface layout

Deliverables:

- KiCad schematic file
- 3D stack-up concept (STEP or layered diagram)
- Pinout allocation sheet

Milestone 3: PCB Layout & Enclosure Fit Check

Duration: 6 Days

Activities:

- Complete PCB layout within 22 × 42 mm and ≤5 mm height
- Place components to minimize thickness and routing complexity
- Export board outline and align with mechanical enclosure
- Conduct design review and DFM validation

Deliverables:

- KiCad PCB layout file
- 3D STEP file of PCB for fit testing
- Gerber and drill files for manufacturing

Milestone 4: Firmware Development & BLE Integration

Duration: 5 Days

Activities:

- Develop core firmware in C/C++ using Nordic SDK
- Implement BLE 5.0 GATT profile
- Add OTA update support and tap gesture logic
- Begin sensor data readout (PPG, IMU, temp)

Deliverables:

- Source code for initial firmware
- BLE service UUIDs and characteristics documentation
- OTA update framework
- Sensor integration test logs

Milestone 5: PCB Manufacturing & Prototype Assembly

Duration: 5 Days

Activities:

- Send final design for PCB manufacturing (JLCPCB or similar)
- Assemble prototype by soldering components
- Connect and test sensors, vibration motor, battery
- Flash firmware and validate bootup & BLE pairing

Deliverables:

- 1–2 fully assembled prototype units
- Assembly checklist and validation report
- Flash image for production testing

Milestone 6: Testing, Debugging & Optimization

Duration: 6 Days

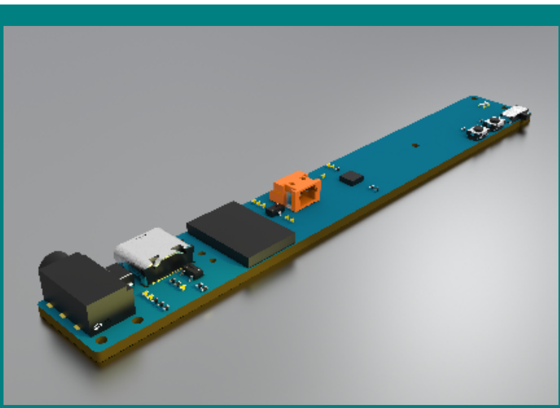
Activities:

- Test biometric functions (HR, SpO_2, temp) under realistic wear
- Validate BLE sync with mobile app
- Optimize power consumption and sleep modes
- Verify pogo pin charging and UART to dock communication

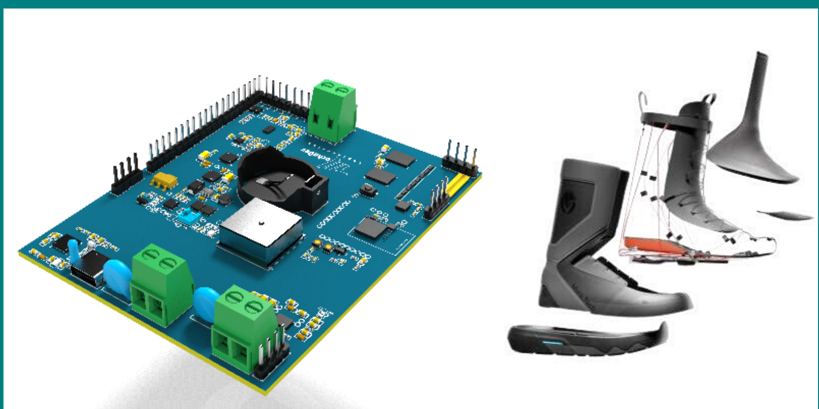
Deliverables:

- Final firmware (with low power optimization
- Tested prototype ready for enclosure
- Full documentation: BOM, test report, firmware guide, BLE profile

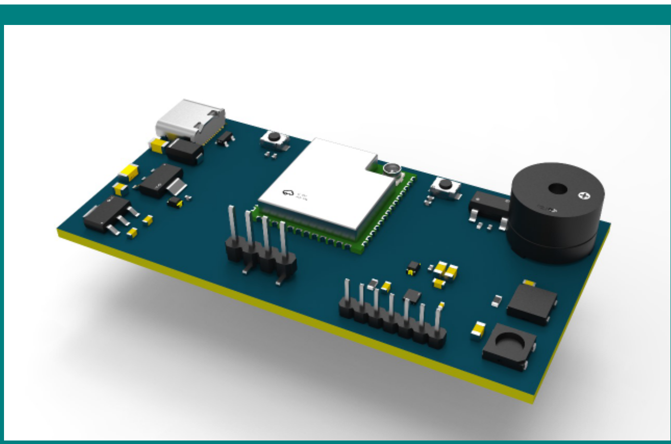
Portfolio



I have designed and developed different wearable devices. One of those includes a pair of smart glasses to assist individuals with mobility impairments in controlling a computer or mobile device's mouse cursor using head movements. This innovative device features a compact multi-layer PCB, integrating a gyro sensor for precise motion tracking and a Bluetooth module for seamless wireless connectivity. The firmware is optimized for real-time responsiveness, ensuring smooth cursor control. Additionally, the system includes a Battery Management System (BMS) to regulate charging, discharging, and temperature, protecting against overcharging or over-discharging. This project enhances accessibility by providing an intuitive and user-friendly solution for individuals with disabilities.



I have designed and developed various wearable devices, including a pair of smart shoes aimed at enhancing performance and comfort through adaptive technology. This innovative footwear features an advanced 8-layer PCB that integrates wireless charging, GPS, and a Bluetooth module for seamless data transmission. Built around the BMD-300 module (nRF52832), the firmware supports real-time biomechanical data capture, specifically optimized for beginner skiers and active users. The system includes an LTC2499 ADC driver for high -resolution pressure sensor readings, programmable RGB lighting, and motor drivers to dynamically adjust the sole's stiffness based on movement patterns. A Battery Management System (BMS) ensures safe power delivery and charging. Data is continuously sent to a mobile app, providing users with interactive feedback and performance analytics to enhance their training experience.



I have also worked on a biomedical sensing project focused on designing and developing an ultra-flexible wearable array for blood oxygen monitoring and subdermal vessel mapping. This innovative device is built around a high-density LED matrix, with red (660nm) and infrared (940nm) LEDs spaced less than 0.3mm apart on an ultra-flexible PCB to conform naturally to the skin's surface. The system uses the AFE4404 analog front-end for precise signal acquisition and processing, enabling accurate SpO_2 measurement and visualization of blood vessel trajectories. Inspired by recent research, this project includes key enhancements in miniaturization and flexibility to improve wearability and sensor performance, aiming to set a new standard in continuous, non-invasive health monitoring.

Client Reviews

PCB and Electrical Design for a Photoreactor

"I enjoy working with this group. They are talented"

Marc Bazin
(USA)

Smart Alarm Clock With E-ink Display

"Fast response time, good planning from the start and good support.

It was a pleasure to work with the team, and I hope to rehire soon again."

Peter Arvad
(Denmark)

Advance Existing Prototype - for Ultra Small Smart Flowmeter

"Muhammad and his team have provided high quality work, downsizing my original PCB substantially and selecting great replacement components for the design requirements. The time of completion for the firmware was overshot more than I had hoped, however, Muhammad has been extremely accommodating, hiring specialized support where needed. My project was not a simple one and I'm certainly pleased with how it has come to life."

Joseph Perez
(USA)

PCB Layout

"Muhammad and his colleagues did an excellent job. From planning to completion, everything went perfectly with my project. Communication was very good, and responses were always quick. Everything was explained clearly, and you could follow the project via Google Docs to see what was being done. If any changes were needed, you could communicate them directly, and they were implemented. They are a highly efficient team that always strives to find a quick solution. I can only recommend the team and would start a project with them again at any time."

Lorenz Stotz

Help Build Supplement Vending Machine Prototype

"Muhammad Bilal and his team are very diligent, creative and hard working. Their ability to be able to problem solve and create solutions for work moving forward has enabled the success of our project now and for the future. I would recommend to anyone."

Brian Hickey
(Ireland)

Thank You!

Thank you for your time and consideration. We would be happy to discuss our proposal further, address any questions you may have, and provide additional insights.

We look forward to your feedback and the opportunity to collaborate on the next steps

Muhammad Bilal Javaid

200+ PROJECT	70+ CLIENT	30+ Countries	98% Response Rate